

References

- Vaibhav Adlakha, Parishad BehnamGhader, Xing Han Lu, Nicholas Meade, and Siva Reddy. Evaluating correctness and faithfulness of instruction-following models for question answering. *arXiv preprint arXiv:2307.16877*, 2023.
- Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, et al. Guidelines for human-ai interaction. In *Proceedings of the 2019 chi conference on human factors in computing systems*, 2019.
- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in ai safety, 2016.
- AutoGPT. Documentation — auto-gpt. <https://docs.agpt.co/>, 2023.
- BabyAGI. Github — babyagi. <https://github.com/yoheinakajima/babyagi>, 2023.
- Carrie J. Cai, Samantha Winter, David F. Steiner, Lauren Wilcox, and Michael Terry. "hello ai": Uncovering the onboarding needs of medical practitioners for human-ai collaborative decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 2019.
- Tianle Cai, Xuezhi Wang, Tengyu Ma, Xinyun Chen, and Denny Zhou. Large language models as tool makers. *arXiv preprint arXiv:2305.17126*, 2023.
- Chroma. Chromadb. <https://github.com/chroma-core/chroma>, 2023.
- Victor Dibia. LIDA: A tool for automatic generation of grammar-agnostic visualizations and infographics using large language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, Toronto, Canada, July 2023. Association for Computational Linguistics.
- Yihong Dong, Xue Jiang, Zhi Jin, and Ge Li. Self-collaboration code generation via chatgpt. *arXiv preprint arXiv:2304.07590*, 2023.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint arXiv:2305.14325*, 2023.
- Atty Eleti, Jeff Harris, and Logan Kilpatrick. Function calling and other api updates. <https://openai.com/blog/function-calling-and-other-api-updates>, 2023.
- Guidance. Guidance. <https://github.com/guidance-ai/guidance>, 2023.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021.
- Sirui Hong, Xiawu Zheng, Jonathan Chen, Yuheng Cheng, Ceyao Zhang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, Liyang Zhou, Chenyu Ran, et al. Metagpt: Meta programming for multi-agent collaborative framework. *arXiv preprint arXiv:2308.00352*, 2023.
- Eric Horvitz. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*, 1999.
- HuggingFace. Transformers agent. https://huggingface.co/docs/transformers/transformers_agents, 2023.
- Geunwoo Kim, Pierre Baldi, and Stephen McAleer. Language models can solve computer tasks. *arXiv preprint arXiv:2303.17491*, 2023.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 2019.

- LangChain. Introduction — langchain. <https://python.langchain.com/en/latest/index.html>, 2023.
- Mike Lewis, Denis Yarats, Yann N Dauphin, Devi Parikh, and Dhruv Batra. Deal or no deal? end-to-end learning for negotiation dialogues. *arXiv preprint arXiv:1706.05125*, 2017.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 2020.
- Beibin Li, Konstantina Mellou, Bo Zhang, Jeevan Pathuri, and Ishai Menache. Large language models for supply chain optimization. *arXiv preprint arXiv:2307.03875*, 2023a.
- Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. Camel: Communicative agents for "mind" exploration of large scale language model society, 2023b.
- Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Zhaopeng Tu, and Shuming Shi. Encouraging divergent thinking in large language models through multi-agent debate, 2023.
- Evan Zheran Liu, Kelvin Guu, Panupong Pasupat, Tianlin Shi, and Percy Liang. Reinforcement learning on web interfaces using workflow-guided exploration. *arXiv preprint arXiv:1802.08802*, 2018.
- Jerry Liu. LlamaIndex, November 2022. URL https://github.com/jerryjliu/llama_index.
- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Alex Graves, Ioannis Antonoglou, Daan Wierstra, and Martin Riedmiller. Playing atari with deep reinforcement learning. *arXiv preprint arXiv:1312.5602*, 2013.
- Roberto Navigli, Simone Conia, and Björn Ross. Biases in large language models: Origins, inventory and discussion. *ACM Journal of Data and Information Quality*, 2023.
- OpenAI. ChatGPT plugins. <https://openai.com/blog/chatgpt-plugins>, 2023.
- Joon Sung Park, Joseph C O’Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. *arXiv preprint arXiv:2304.03442*, 2023.
- Md Rizwan Parvez, Wasi Uddin Ahmad, Saikat Chakraborty, Baishakhi Ray, and Kai-Wei Chang. Retrieval augmented code generation and summarization. *arXiv preprint arXiv:2108.11601*, 2021.
- Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. Gorilla: Large language model connected with massive apis. *arXiv preprint arXiv:2305.15334*, 2023.
- Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 11 2019. URL <https://arxiv.org/abs/1908.10084>.
- Semantic-Kernel. Semantic kernel. <https://github.com/microsoft/semantic-kernel>, 2023.
- Bokui Shen, Fei Xia, Chengshu Li, Roberto Martín-Martín, Linxi Fan, Guanzhi Wang, Claudia Pérez-D’Arpino, Shyamal Buch, Sanjana Srivastava, Lyne Tchapmi, et al. igibson 1.0: A simulation environment for interactive tasks in large realistic scenes. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2021.
- Tianlin Shi, Andrej Karpathy, Linxi Fan, Jonathan Hernandez, and Percy Liang. World of bits: An open-domain platform for web-based agents. In *International Conference on Machine Learning*. PMLR, 2017.

- Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. ALFWorld: Aligning Text and Embodied Environments for Interactive Learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021. URL <https://arxiv.org/abs/2010.03768>.
- Oriol Vinyals, Timo Ewalds, Sergey Bartunov, Petko Georgiev, Alexander Sasha Vezhnevets, Michelle Yeo, Alireza Makhzani, Heinrich Küttler, John Agapiou, Julian Schrittwieser, et al. Starcraft ii: A new challenge for reinforcement learning. *arXiv preprint arXiv:1708.04782*, 2017.
- Chi Wang, Qingyun Wu, Markus Weimer, and Erkang Zhu. Flaml: A fast and lightweight autotml library. *Proceedings of Machine Learning and Systems*, 2021.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023a.
- Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, et al. A survey on large language model based autonomous agents. *arXiv preprint arXiv:2308.11432*, 2023b.
- Daniel S. Weld and Oren Etzioni. The first law of robotics (a call to arms). In *AAAI Conference on Artificial Intelligence*, 1994.
- Max Woolf. Langchain problem. <https://minimaxir.com/2023/07/langchain-problem/>, 2023.
- Yiran Wu, Feiran Jia, Shaokun Zhang, Qingyun Wu, Hangyu Li, Erkang Zhu, Yue Wang, Yin Tat Lee, Richard Peng, and Chi Wang. An empirical study on challenging math problem solving with gpt-4. *arXiv preprint arXiv:2306.01337*, 2023.
- Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. The rise and potential of large language model based agents: A survey. *arXiv preprint arXiv:2309.07864*, 2023.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.

A Related Work

We examine existing LLM-based agent systems or frameworks that can be used to build LLM applications. We categorize the related work into single-agent and multi-agent systems and specifically provide a summary of differentiators comparing AutoGen with existing multi-agent systems in Table 1. Note that many of these systems are evolving open-source projects, so the remarks and statements about them may only be accurate as of the time of writing. We refer interested readers to detailed LLM-based agent surveys (Xi et al., 2023; Wang et al., 2023b)

Single-Agent Systems:

- **AutoGPT:** AutoGPT is an open-source implementation of an AI agent that attempts to autonomously achieve a given goal (AutoGPT, 2023). It follows a single-agent paradigm in which it augments the AI model with many useful tools, and does not support multi-agent collaboration.
- **ChatGPT+ (with code interpreter or plugin):** ChatGPT, a conversational AI service or agent, can now be used alongside a code interpreter or plugin (currently available only under the premium subscription plan ChatGPT Plus) (OpenAI, 2023). The code interpreter enables ChatGPT to execute code, while the plugin enhances ChatGPT with a wide range of curated tools.
- **LangChain Agents:** LangChain is a general framework for developing LLM-based applications (LangChain, 2023). LangChain Agents is a subpackage for using an LLM to choose a sequence of actions. There are various types of agents in LangChain Agents, with the ReAct agent being a notable example that combines reasoning and acting when using LLMs (mainly designed for LLMs prior to ChatGPT) (Yao et al., 2022). All agents provided in LangChain Agents follow a single-agent paradigm and are not inherently designed for communicative and collaborative modes. A significant summary of its limitations can be found in (Woolf, 2023). Due to these limitations, even the multi-agent systems in LangChain (e.g., re-implementation of CAMEL) are not based on LangChain Agents but are implemented from scratch. Their connection to LangChain lies in the use of basic orchestration modules provided by LangChain, such as AI models wrapped by LangChain and the corresponding interface.
- **Transformers Agent:** Transformers Agent (HuggingFace, 2023) is an experimental natural-language API built on the transformers repository. It includes a set of curated tools and an agent to interpret natural language and use these tools. Similar to AutoGPT, it follows a single-agent paradigm and does not support agent collaboration.

AutoGen differs from the single-agent systems above by supporting multi-agent LLM applications.

Multi-Agent Systems:

- **BabyAGI:** BabyAGI (BabyAGI, 2023) is an example implementation of an AI-powered task management system in a Python script. In this implemented system, multiple LLM-based agents are used. For example, there is an agent for creating new tasks based on the objective and the result of the previous task, an agent for prioritizing the task list, and an agent for completing tasks/sub-tasks. As a multi-agent system, BabyAGI adopts a static agent conversation pattern, i.e., a predefined order of agent communication, while AutoGen supports both static and dynamic conversation patterns and additionally supports tool usage and human involvement.
- **CAMEL:** CAMEL (Li et al., 2023b) is a communicative agent framework. It demonstrates how role playing can be used to let chat agents communicate with each other for task completion. It also records agent conversations for behavior analysis and capability understanding. An Inception-prompting technique is used to achieve autonomous cooperation between agents. Unlike AutoGen, CAMEL does not natively support tool usage, such as code execution. Although it is proposed as an infrastructure for multi-agent conversation, it only supports static conversation patterns, while AutoGen additionally supports dynamic conversation patterns.
- **Multi-Agent Debate:** Two recent works investigate and show that multi-agent debate is an effective way to encourage divergent thinking in LLMs (Liang et al., 2023) and to improve the factuality and reasoning of LLMs (Du et al., 2023). In both works, multiple LLM inference instances are constructed as multiple agents to solve problems with agent debate. Each agent is simply an LLM inference instance, while no tool or human is involved, and the inter-agent conversation needs to follow a pre-defined order. These works attempt to build LLM applications with multi-agent conversation, while AutoGen, designed as a generic infrastructure, can be used to facilitate this development and enable more applications with dynamic conversation patterns.

- **MetaGPT**: MetaGPT (Hong et al., 2023) is a specialized LLM application based on a multi-agent conversation framework for automatic software development. They assign different roles to GPTs to collaboratively develop software. They differ from AutoGen by being specialized solutions to a certain scenario, while AutoGen is a generic infrastructure to facilitate building applications for various scenarios.

There are a few other specialized single-agent or multi-agent systems, such as Voyager (Wang et al., 2023a) and Generative Agents (Park et al., 2023), which we skip due to lower relevance. In Table 1, we summarize differences between AutoGen and the most relevant multi-agent systems.

Table 1: Summary of differences between AutoGen and other related multi-agent systems. **infrastructure**: whether the system is designed as a generic infrastructure for building LLM applications. **conversation pattern**: the types of patterns supported by the implemented systems. Under the ‘static’ pattern, agent topology remains unchanged regardless of different inputs. AutoGen allows flexible conversation patterns, including both static and dynamic patterns that can be customized based on different application needs. **execution-capable**: whether the system can execute LLM-generated code; **human involvement**: whether (and how) the system allows human participation during the execution process of the system. AutoGen allows flexible human involvement in multi-agent conversation with the option for humans to skip providing inputs.

Aspect	AutoGen	Multi-agent Debate	CAMEL	BabyAGI	MetaGPT
Infrastructure	✓	✗	✓	✗	✗
Conversation pattern	flexible	static	static	static	static
Execution-capable	✓	✗	✗	✗	✓
Human involvement	chat/skip	✗	✗	✗	✗